

A temporally restricted function of the Dopamine receptor Dop1R2 during memory formation

Reviewed Preprint

v1 • July 30, 2024

Not revised

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Abstract

Summary

Dopamine is a crucial neuromodulator, which is involved in many brain processes, including learning and the formation of memories. Dopamine acts through multiple receptors and controls an intricate signaling network to regulate different tasks. While the diverse functions of dopamine are intensely studied, the interplay and role of the distinct dopamine receptors to regulate different processes is less well understood. An interesting candidate is the dopamine receptor Dop1R2 (also known as Damb), as it could connect to different downstream pathways. Dop1R2 is reported to be involved in forgetting and memory maintenance, however, the circuits requiring the receptors are unknown. To study Dop1R2 and its role in specific spatial and temporal contexts, we generated a conditional knock-out line using the CRISPR-Cas9 technique. Two FRT sites were inserted, allowing flippase-mediated excision of the dopamine receptor in neurons of interest. To study the function of Dop1R2, we knocked it out conditionally in the Mushroom body of *Drosophila melanogaster*, a well-studied brain region for memory formation. We show that Dop1R2 is required for later memory forms but not for short-term memories for both aversive and appetitive memories. Moreover, Dop1R2 is specifically required in the α/β -lobe and the α'/β' -lobe but not in the γ -lobe of the Mushroom body. Our findings show a spatially and temporally restricted role of Dop1R2 in the process of memory formation highlighting the differential requirement of receptors during distinct phases of learning.

eLife assessment

The authors design and implement an elegant strategy to delete genomic sequences encoding the dopamine receptor dop1R2 from specific subsets of mushroom body neurons (ab, a'b' and gamma) and show that while none of these manipulations affect short term appetitive or aversive memory, loss of dop1R2 from ab or a'b' block the ability of flies to display measurable forms of longer forms of appetitive memory. These findings are **valuable** in confirming and/or moderating prior observations, with better genetic perturbation techniques and **convincing** data to support the authors' main conclusions.

<https://doi.org/10.7554/eLife.99368.1.sa3>

Introduction

The neuromodulator dopamine is involved in a plethora of brain functions. Among them are learning and memory, reinforcement signaling, reward, arousal, and motor functions (Klein et al., 2019 [DOI](#); Missale et al., 1998 [DOI](#); Tritsch & Sabatini, 2012 [DOI](#)). Perturbations in the dopaminergic system are associated with diseases like Parkinson's disease, addiction, Depression, Schizophrenia, and many more (Klein et al., 2019 [DOI](#); Missale et al., 1998 [DOI](#); Tritsch & Sabatini, 2012 [DOI](#)). Therefore, it is important to understand dopamine signaling in greater detail allowing to develop more effective therapeutics or preventive measures.

Dopamine is synthesized from tyrosine in dopaminergic neurons and binds to G-protein coupled receptors (GPCRs) (Klein et al., 2019 [DOI](#); Missale et al., 1998 [DOI](#); Yamamoto & Seto, 2014 [DOI](#)). These dopamine receptors have seven transmembrane domains, with an intracellular C-terminus and an extracellular N-terminus. The receptors interact with different G-Proteins, which are heterotrimeric protein complexes consisting of an α -, β -, and γ -subunit. The G-protein complex binds to the third intracellular loop and the C-terminus of the receptors. Humans and other mammals express five different dopamine receptors, separated into two types. Type 1 dopamine receptors elevate protein kinase A (PKA) signaling and cyclic adenosine monophosphate (cAMP) levels via the alpha subunit G_{α_s} , whereas type 2 receptors inhibit PKA activity via $G_{\alpha_{i/o}}$ thus reducing cAMP levels (Klein et al., 2019 [DOI](#); Missale et al., 1998 [DOI](#); Tritsch & Sabatini, 2012 [DOI](#)). Downstream of cAMP and PKA, transcriptional activation mediated by c-AMP response element-binding protein leads to the expression of immediate early genes as a response to dopamine signaling (Carlezon et al., 2005 [DOI](#); Neves et al., 2002 [DOI](#)). This pathway is a key regulator of long-term memory (LTM) formation (Alberini & Kandel, 2014 [DOI](#); Kaldun & Sprecher, 2019 [DOI](#); Kandel, 2012 [DOI](#)). Moreover, Dopamine receptors are able to modulate internal Ca^{2+} levels by engaging the Phospholipase C signaling pathway. Furthermore, the G-Protein β and γ subunits can directly modulate voltage-gated and ligand-gated ion channels (Klein et al., 2019 [DOI](#); Ledonne & Mercuri, 2017 [DOI](#); Missale et al., 1998 [DOI](#); Nishi et al., 2011 [DOI](#); Tritsch & Sabatini, 2012 [DOI](#)). However, the coordination of the receptors and the different dopamine mediated processes in specific circuits is still a mystery. Therefore, an accessible and well-characterized model is required, like the olfactory circuit of *Drosophila melanogaster*. The connectome of this circuit is known, and it can be manipulated with the sophisticated genetic tools of the fly [Takemura, 2017 #383; Li, 2020 #1015] [Owald, 2015 #200]. Similar to mammals, the fly uses dopamine in learning, memory, forgetting, negative and positive reinforcement, locomotion, and sleep and arousal regulation (Berry et al., 2012 [DOI](#); Burke et al., 2012 [DOI](#); Karam et al., 2020 [DOI](#); Sabandal et al., 2021 [DOI](#); Sabandal et al., 2020 [DOI](#); Siju et al., 2021 [DOI](#); Siju et al., 2020 [DOI](#); Sitaraman et al., 2015 [DOI](#); Tomita et al., 2017 [DOI](#); Waddell, 2013 [DOI](#); Yamamoto & Seto, 2014 [DOI](#)). Additionally, L-Dopa, the precursor for dopamine and a powerful receptor agonist, as well as other pharmaceuticals have been shown to function in the fly (Yamamoto & Seto, 2014 [DOI](#)). Thus, the dopaminergic system can be studied in *Drosophila*, taking advantage of the available genetic tools.

The fly uses four different dopamine receptors. Dop1R1 (Dumb) (Kim et al., 2003 [DOI](#); Sugamori et al., 1995 [DOI](#)) and Dop1R2 (Damb) (Feng et al., 1996 [DOI](#); Han et al., 1996 [DOI](#)) are type 1 receptors. Dop2R is a type 2 receptor (Hearn et al., 2002 [DOI](#)), and dopEcR is a type 1 dopamine receptor that also uses ecdysone as a ligand (Srivastava et al., 2005 [DOI](#)). All four receptors were shown to be involved in learning, memory and forgetting (Berry et al., 2012 [DOI](#); Karam et al., 2020 [DOI](#); Kim et al., 2007 [DOI](#); Lark et al., 2017 [DOI](#); Qi & Lee, 2014 [DOI](#); Qin et al., 2012 [DOI](#); Scholz-Kornehl & Schwarzel, 2016 [DOI](#); Sun et al., 2020 [DOI](#); Zhou et al., 2019 [DOI](#)). While it is well established that Dop1R1 is crucial for learning and short-term memory (Kim et al., 2007 [DOI](#); Qin et al., 2012 [DOI](#)), the role of the other receptors is less clear. Dop1R2 is an interesting candidate to study, due to its ability to couple to two different G-proteins, G_{α_s} , which engages the cAMP pathway and G_{α_q} , which couples to Ca^{2+} -signaling (Han et al., 1996 [DOI](#); Himmelmreich et al., 2017 [DOI](#); Sun et al., 2020 [DOI](#)). This could allow

Dop1R2 to modulate learning and memory in a complex fashion. The receptor is mainly expressed in the Mushroom body (MB) (Crocker et al., 2016 [↗](#); Croset et al., 2018 [↗](#); Han et al., 1996 [↗](#); Kim et al., 2007 [↗](#); Lark et al., 2017 [↗](#)), an important brain region for olfactory associative learning (Aso, Hattori, et al., 2014; Aso, Sitaraman, et al., 2014; Cognigni et al., 2018 [↗](#)). Work on a mutant line suggests that Dop1R2 is involved in forgetting (Berry et al., 2012 [↗](#)). However, a study using an RNAi-line suggests that the receptor plays a role in memory maintenance (Sun et al., 2020 [↗](#)). As these studies used different learning assays – aversive and appetitive respectively as well as different methods, it is unclear if Dop1R2 has different functions for the different reinforcement stimulus. To resolve this problem, we generated a transgenic line to conditionally knock down Dop1R2, in a spatial and temporal specific manner. Using CRISPR-Cas9 and homology-directed repair (HDR), we included FRT sites in the endogenous locus of *Dop1R2* for flippase-mediated excision. In addition, we inserted an HA-Tag, to monitor the spatial localization of the receptor. This also allows us to visualize the efficiency of the flip-out. We used this line to study the role of the receptor in learning and memory in the MB for both aversive and appetitive conditioning. Upon flip out of the receptor in the MB, 2h memory and long-term memory (LTM) are impaired. Similar results are obtained when we flip out Dop1R2 specifically in the α/β -lobes and α'/β' -lobes of the MB, which are involved in those memory phases. Therefore, Dop1R2 is required in the α/β -lobes and α'/β' -lobes for later memory forms.

Results

Generation of the Dop1R2 conditional knock-out line

To be able to study both the requirement of Dop1R2 in specific neurons as well as the localization of the receptor we generated a transgenic line that allows defined spatial and temporal knock-out of the receptor. In short, we inserted two FRT sites for flippase-mediated excision (Golic & Lindquist, 1989 [↗](#); Gratz et al., 2014 [↗](#)) as well as a 3xHA-tag to study the localization of the receptor. The Ha-Tag was chosen to minimize interference with the receptor structure. A representation of the receptor is depicted in **Figure 1A** [↗](#). The 3rd intracellular loop between transmembrane domains (TMDs) five and six as well as the C-terminal tail are required for G-protein binding (Missale et al., 1998 [↗](#)). The position of the FRT sites and the HA-tag is indicated. **Figure 1B** [↗](#) gives an overview of the strategy. The endogenous locus is cut twice by CRISPR-Cas9 to replace it with the Dop1R2 sequence carrying the FRT sites and the HA-Tag, using a donor plasmid as template for homolog-directed repair. The donor plasmid also contains the region directly upstream and downstream of the exchange site as homology arms to align the plasmid. The plasmid carrying the sequence for the two gRNAs as well as the donor plasmid were injected into embryos with maternal Cas9 expression (*nos>Cas9*). Using the Gal4-UAS system to express flippase in neurons of interest the receptor can be flipped out in those neurons while keeping it functional in the rest of the brain. The first FRT site was placed in the intron before the first coding exon, which contains all seven TMDs (**Figure 1C** [↗](#)). Since Dop1R2 has three transcript isoforms with differences in the C-terminus the second FRT site was placed at the end of the last common exon. Thus, FLP-mediated recombination will lead to deletion of the two common coding exons including all TMDs. Successful insertion was verified by sequencing.

Dop1R2^{cko} is in the Mushroom body

To monitor the localization of the tagged dopamine receptors we first stained brains with an antibody against the HA-Tag in one-week-old flies. In the control line (**Figure 1D** [↗](#)) the HA-tag does not show any specific staining. In the *dop1R2^{cko}* line, we see a clear signal in the entire MB (**Figure 1E** [↗](#)). This matches previous reports as well as single-cell RNAseq data showing that *Dop1R2* is expressed in the MB (Croset et al., 2018 [↗](#); Han et al., 1996 [↗](#)).

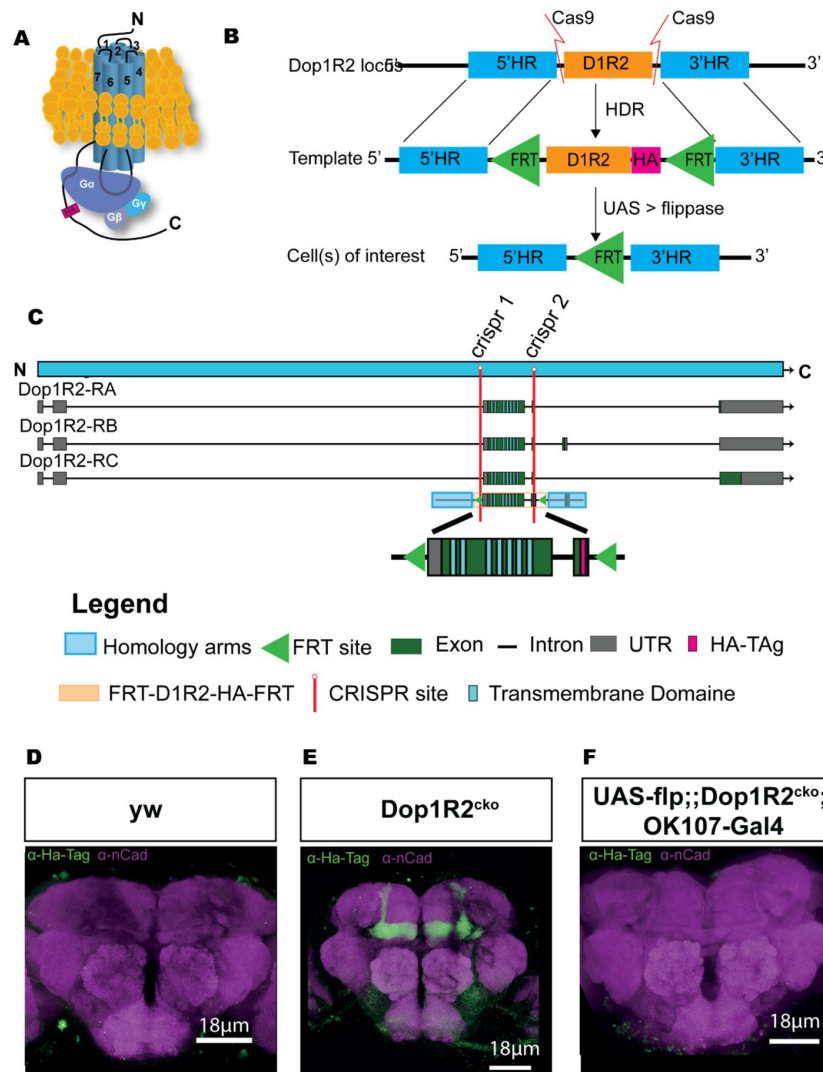


Fig 1.

Generation of a conditional knockout allele for Dop1R2.

A) Schematic representation of the receptor structure and interaction with the G-Protein. The FRT sites and HA-Tag are indicated B) Schematic representation of the conditional knock-out system. The endogenous *Dop1R2* was replaced using CRISPR-Cas9 mediated homology-directed repair (HDR) from a donor plasmid. The plasmid contained the common coding exons of *Dop1R2* with an HA-Tag in the C-Terminus and two FRT sites flanking this sequence. In the resulting *Dop1R2*^{cko} allele the inserted HA-Tag and *Dop1R2* sequence can be removed by flippase (FLP) recombinase in cells of interest. C) Schematic representation of the *Dop1R2* gene locus with 3 different transcript isoforms. The location of the two used CRISPR sites are highlighted in red. The positions of the transmembrane domains in the isoforms and in the donor plasmid, are indicated. D-F) Dop1R2::HA expression in a frontal brain confocal section of D yw, E Dop1R2^{cko} or F UAS-flp;; dop1R2^{cko}, OK107-Gal4 flies aged one week. The HA-Tag was visualized using an anti-HA-Tag antibody (green). Brain structures were labeled with anti-N-cadherin (nCad, magenta) antibodies. Scale bar: 18 μm.

After flipping the receptor out in the MB using the MB-specific OK107-Gal4 driver in combination with UAS-flp abolishes the signal, demonstrating the efficiency of the FRT sites (**Figure 1F** [↗](#)). Therefore, the flip-out system seems to work as planned.

Short-term memory is not affected by the loss of Dop1R2

We wanted to see if flipping out Dop1R2 in the MB affects memory acquisition and STM by using classical olfactory conditioning. In short, a group of flies is presented with an odor coupled to an electric shock (aversive) or sugar (appetitive) followed by a second odor without stimulus. For assessing memory flies can freely choose between the odors either directly after training (STM) or at a later timepoint.

To ensure that the introduced genetic changes to the Dop1R2 locus do not interfere with behavior we first checked the sensory responses of that line (Sup A-D). The Dop1R2^{cko} line shows comparable odor responses (Sup A+B) as well as sugar and shock response (Sup C+D) to the control line. Aversive STM (Sup E), as well as aversive 2h memory (Sup F), are not significantly different from the control line. Moreover, appetitive STM is also comparable to the control (Sup G). Thus, the introduced changes to dop1R2 do not interfere with normal receptor function.

To test the requirement of dop1R2 for short-term memory we assessed memory performance directly after training for flies with flipped-out dop1R2 in the whole MB or in individual lobes along with the parental controls. The flies were aged for a week before undergoing classical olfactory conditioning. Both aversive STM and appetitive STM were tested. But first, we ensured that the whole MB flip-out line responds normally to the used odors and stimuli. Flp/+;; Dop1R2^{cko}; OK107-Gal4/+ flies show comparable responses as the parental controls to both odors (Sup H+I), shock (Sup J) and sugar (Sup K).

For aversive STM flipping out Dop1R2 in the whole MB does not change STM compared to parental controls (**Figure 2A** [↗](#)). Next, we flipped Dop1R2 in the γ -lobe using 5HTR1B-Gal4. This MB lobe is involved in STM (Blum et al., 2009 [↗](#); Trannoy et al., 2011 [↗](#); Zars et al., 2000 [↗](#)). Matching the results for the whole MB flip out, we do not see a change in the performance score (**Figure 2B** [↗](#)). Similar results are obtained when we flip the receptor in the α/β -lobes using c739-Gal4 (**Figure 2C** [↗](#)) or α'/β' -lobes using c305a-Gal4 (**Figure 2D** [↗](#)).

Reward STM is also not changed when Dop1R2 is flipped out in the whole MB (**Figure 2E** [↗](#)) in the γ -lobe (**Figure 2F** [↗](#)), the α/β -lobes (**Figure 2G** [↗](#)) or the α'/β' -lobes (**Figure 2H** [↗](#)). Taken together the results indicate that Dop1R2 is not required for STM in the MB lobes.

2h memory is impaired by loss of Dop1R2

As Dop1R2 was previously described to be involved in forgetting and/or memory maintenance we wanted to assess later time points after training.

First, we looked at two hours after aversive training. Flip-out in the whole Mushroom body leads to a reduced performance score (**Figure 3A** [↗](#)). Next, we wondered which MB lobe might cause this reduction. Therefore, we flipped out Dop1R2 in γ -lobe using 5HTR1B-Gal4, in the α/β -lobes using c739-Gal4 or α'/β' -lobes using c305a-Gal4. Loss of Dop1R2 in the γ -lobe does not reduce the memory performance (**Figure 3B** [↗](#)).

However, loss of Dop1R2 in both the α/β -lobes (**Figure 3C** [↗](#)) or the α'/β' -lobes (**Figure 3D** [↗](#)) impaired 2h memory after aversive training.

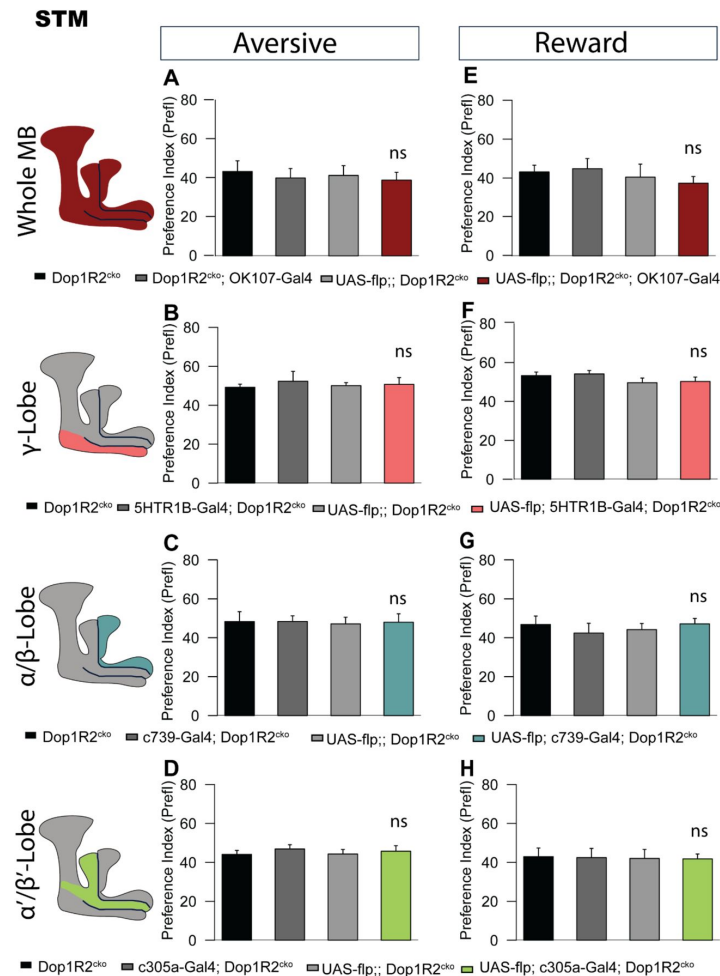


Fig 2.

Short-term memory of flies with knock-out of Dop1R2 in the mushroom Body.

A-D) Aversive training, E-H) reward training. A) and E) Whole MB flip-out using OK107-Gal4 and parental controls. B) and F) γ-lobe flip-out using 5HTR1B-Gal4 and parental controls. C) and G) α/β-lobe flip-out using c739-Gal4 and parental controls. D) and H) α'/β'-lobe flip-out using c305a-Gal4 and parental controls. No performance impairment was observed in any of the tested conditions. See S2 for sensory controls and S1 Table for the data. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: p > 0.05) determined by ANOVA Tukey HSD.

We tested the same for reward memory using sugar as reinforcement. Flip out of Dop1R2 in the whole MB (**Figure 3E**) results in a reduced performance score. Loss of Dop1R2 in the γ -lobe does not affect memory performance (**Figure 3F**). As for aversive training, flip out of Dop1R2 in the α/β -lobes (**Figure 3G**) or the α'/β' -lobes (**Figure 3H**) impaired 2h memory after reward training. However, the reduction is not as severe as for aversive training. Taken together the results indicate that Dop1R2 is required for aversive 2h memory as well as reward 2h memory in the α/β -lobes and the α'/β' -lobes but is dispensable in the γ -lobe.

24h memory is impaired by loss of Dop1R2

Next, we wanted to see if later memory forms are also affected. One cycle of reward training is sufficient to create LTM (Krashes & Waddell, 2008), so we looked at 24h appetitive memory.

Flipping out *dop1R2* in the whole MB causes a reduced 24h memory performance (**Figure 4A**). No phenotype was observed when *dop1R2* was flipped out in the γ -lobe (**Figure 4B**). However, similar to 2h memory, loss of *dop1R2* in the α/β -lobes (**Figure 4C**) or the α'/β' -lobes (**Figure 4D**) causes a further reduction in memory performance. Thus, *dop1R2* seems to be involved in appetitive LTM in the α/β -lobes and the α'/β' -lobes.

Discussion

We have generated a conditional knock-out line for the dopamine receptor Dop1R2 following a similar approach as in Widmer et al. 2018 (Widmer et al., 2018). To achieve this, FRT sites were inserted in front of the start codon and in the C-Terminus of Dop1R2 using CRISPR-Cas9 mediated homology-directed repair. In addition, an HA-Tag was inserted to be able to visualize the dopamine receptor expression as well as verify successful flip out. Using an anti-HA-Tag antibody we were able to visualize Dop1R2 in the MB of the generated line. This matches previous reports for the expression of *Dop1R2* (Crocker et al., 2016; Croset et al., 2018; Han et al., 1996; Kim et al., 2007; Sun et al., 2020). Moreover, upon knocking out Dop1R2 with an MB-specific driver, the HA-tag labeling disappears indicating that the conditional knock-out system works. The HA-Tag could also be useful to study the localization of Dop1R2 within the MB lobes, for example if it is close to synapses of dopaminergic neurons.

To get a better overview of Dop1R2's role in the Mushroom body we analyzed appetitive memory at different timepoints after training in the individual MB lobes. Loss of *Dop1R2* in the whole MB as well as the α/β -lobe or the α'/β' -lobe impairs 24h reward memory. This observation matches previous studies. Using *Dop1R2-RNAi* in the MB, Sun et al. (Sun et al., 2020) showed, that STM is intact, while LTM is impaired. When knocking down the Raf/MAPK pathway they get a similar phenotype. Moreover, expression of a constitutive active *Raf* allele rescues the Dop1R2 dependent LTM deficit and Dop1R2 seems to be required for the phosphorylation of the MAPK. Therefore, Dop1R2 dependent activation of the Raf/MAPK pathway is required for stabilization of reward LTM memory. Another study shows that reward LTM requires food with a high energetic value (Musso et al., 2015). This signal seems to be relayed by the dopaminergic neuron (DAN) MB-MP1, that showed temporally restricted oscillating activity early post-training (Musso et al., 2015; Pavlovsky et al., 2018). Furthermore, knock-down of Dop1R2 using RNAi impaired reward LTM while leaving STM intact. Therefore, dopaminergic signaling through MB-MP-1 and Dop1R2 could indicate the energetic value of the reward and decide if LTM should be formed or not. Interestingly, the MB-MP1 DAN arborizes in the spur of the γ -lobe as well as the inner core of the pedunculus, which consists of the axons of the α/β Kenyon Cells (KCs) (Tanaka et al., 2008), which according to our results require Dop1R2 for reward LTM. Loss of Dop1R2 in the Mushroom Body Output neuron (neuron) MVP2 also impaired reward LTM while leaving STM intact (Pavlovsky et al., 2018). This GABAergic MVP2 MBON forms a feedback circuit with the

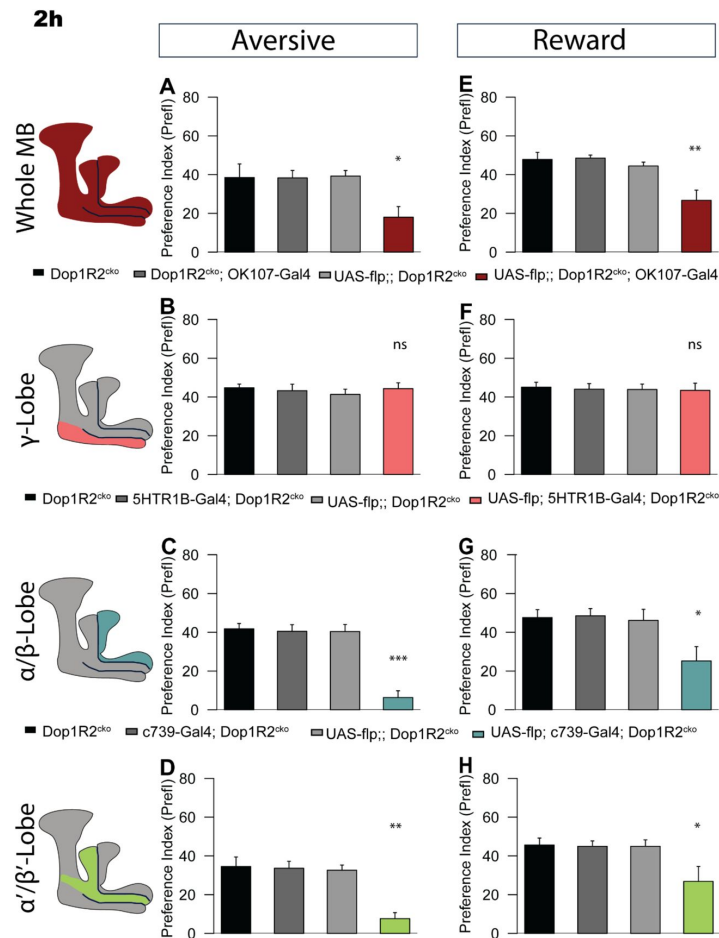


Fig 3.

2h memory of flies with knock-out of Dop1R2 in the mushroom Body.

A-D) Aversive training, E-H) reward training. A) and E) Whole MB flip-out using OK107-Gal4 and parental controls. B) and F) γ-lobe flip-out using 5HTR1B-Gal4 and parental controls. C) and G) α/β-lobe flip-out using c739-Gal4 and parental controls. D) and H) α'/β'-lobe flip-out using c305a-Gal4 and parental controls. For whole MB flip-out, α/β-lobes and α'/β'-lobes both aversive and appetitive 2h memory performance is impaired. Loss of Dop1R2 in the γ-lobe does not affect 2h memory. See S2 for sensory controls and S1 Table for the data. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: p > 0.05) determined by ANOVA Tukey HSD.

LTM

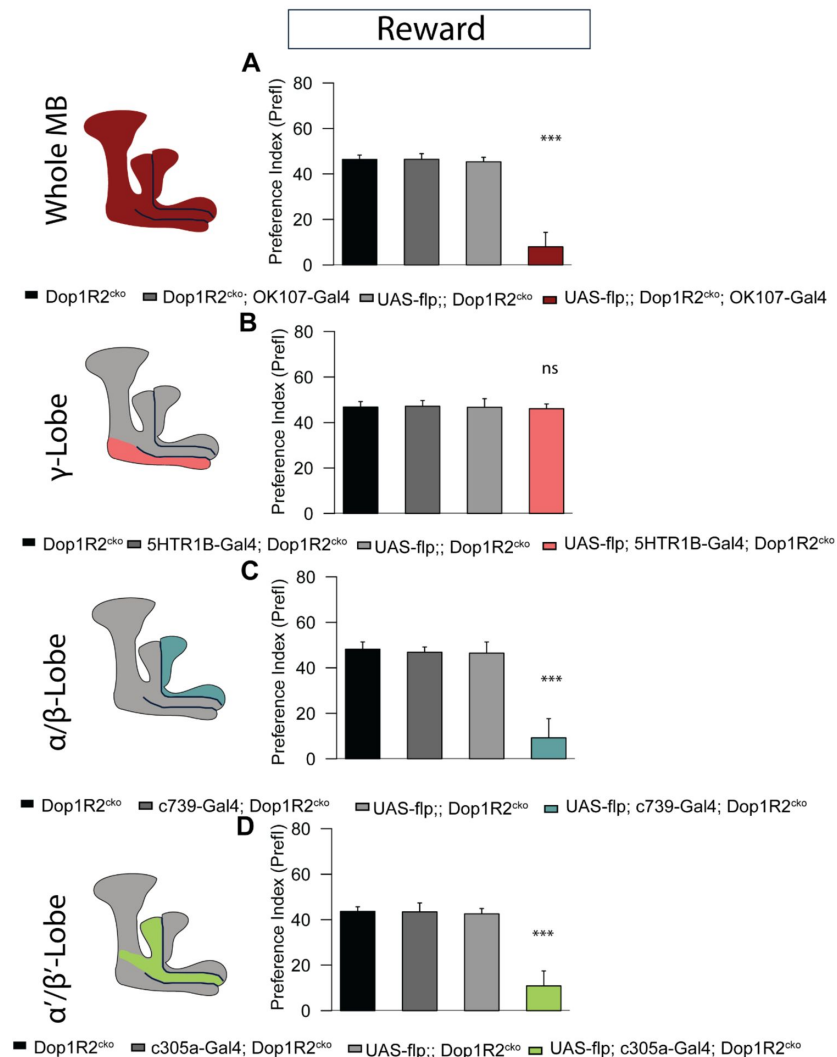


Fig 4.

24h reward memory of flies with knock-out of Dop1R2 in the mushroom Body.

A) Whole MB flip-out using OK107-Gal4 and parental controls. B) γ-lobe flip-out using 5HTR1B-Gal4 and parental controls. C) α/β-lobe flip-out using c739-Gal4 and parental controls. D) α'/β'-lobe flip-out using c305a-Gal4 and parental controls. For whole MB flip-out, α/β-lobes and α'/β'-lobes appetitive 24h memory performance is impaired. Loss of Dop1R2 in the γ-lobe does not affect 24h memory. See S2 for sensory controls and S1 Table for the data. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: p > 0.05) determined by ANOVA Tukey HSD.

MB-MP1 DAN. After training, the oscillatory activity of MB-MP1 is enhanced, while MVP2 is inhibited. After 30min, MVP2 gets activated and MB-MP1 is inhibited. Moreover, Dop1R2 seems to be required for modulating this feedback loop.

The MBONs are an important contributor to memory. Moreover, MBONs receive dopaminergic input and seem to express dopamine receptors (Crocker et al., 2016 [↗](#)). We showed that Dop1R2 is required in the α/β -lobe and the α'/β' -lobe for reward LTM formation. Interestingly, MBONs which are involved in reward LTM (Ichinose et al., 2015 [↗](#); Oswald et al., 2015 [↗](#); Plaçais et al., 2013 [↗](#)), have arborization in the α/β -lobe and the α'/β' -lobe. Thus, Dop1R2 could modulate these connections as well.

All in all, Dop1R2 is required for reward LTM formation and loss of the receptor impairs LTM. Dop1R2 seems to influence reward LTM in different ways. Firstly, by acting on the Raf-MAPK pathway to stabilize memory, secondly by relaying the energetic content of the reward and third, by modulating the MB-MP1-MVP2 loop.

For aversive conditioning, we observe that loss of Dop1R2 in the MB leads to impaired 2h memory, whereas STM is intact. Moreover, Dop1R2 seems to be required in the α/β -lobe and the α'/β' -lobe. A previous study using a mutant for Dop1R2, that also affects the neighboring gene GC1907, observed higher memory retention (Berry et al., 2012 [↗](#)). Further, lack of Dop1R2 impairs reversal learning. It is proposed that Dop1R2 acts on the RAC-forgetting pathway (Cervantes-Sandoval et al., 2016 [↗](#); Shuai et al., 2010 [↗](#)). Interestingly, Dop1R2 can act on two different downstream second messenger systems by using two different G-Proteins (Himmelreich et al., 2017 [↗](#)). By coupling to $G_{\alpha s}$ the cAMP pathway is activated. By coupling to $G_{\alpha q}$ the Ca^{2+} messenger system is activated. Furthermore, knocking down $G_{\alpha q}$ pan-neuronally or in the MB leads to memory enhancement 3h after training, but not 6h after training (Himmelreich et al., 2017 [↗](#)). Thus, Dop1R2 could regulate the RAC forgetting pathway through $G_{\alpha q}$. However, also other receptors could use $G_{\alpha q}$ and mediate forgetting.

Berry et al. also observed that blocking output of DANs inhibits forgetting, while activating DANs accelerates forgetting (Berry et al., 2012 [↗](#)). This modulation seems to require ongoing activity of the MP1 DAN together with further DANs. Interestingly, using a spaced training protocol and a Dop1R2 line another study showed impaired LTM (Plaçais et al., 2017 [↗](#)). Furthermore, after spaced training, flies have a higher energy uptake and the energy metabolism is upregulated in the MB. This increase in energy consumption is mediated by dopaminergic signaling from the MB-MP1 DAN. Loss of Dop1R2 in the MB abolishes the increase in energy consumption. Therefore, Dop1R2 seems to be important for aversive LTM formation by regulating energy consumption. Thus, like for reward LTM formation, aversive LTM seems to require sufficient energy. Starved flies reduce the formation of aversive LTM (Plaçais & Preat, 2013 [↗](#); Plaçais et al., 2012 [↗](#)). This information seems to be relayed through ongoing oscillation of MB-MP1 and Dop1R2 after training. In addition, the MVP2 MBON might also be involved (Ueoka et al., 2017 [↗](#)). Therefore, the gating mechanism in both aversive and reward LTM formation seems to require Dop1R2 (Pavlovsky et al., 2018 [↗](#)). In addition, the MAPK signaling pathway might also be required in aversive LTM formation by activating transcription factors like CREB and c-fos (Miyashita et al., 2018 [↗](#)).

Both Dop1R2 and the ongoing activity of MB-MP1 seem to have multiple roles directly after training in a short time window (Berry et al., 2012 [↗](#); Plaçais et al., 2017 [↗](#); Plaçais et al., 2012 [↗](#)). The circuit acts like a gating mechanism for LTM to ensure, that there is enough energy for continuing LTM formation. The MBON MVP2 acts as feedback loop to regulate the activity of the DAN. Moreover, both the RAC and the Raf/MAPK signaling pathway seem to be engaged to either forget or stabilize the memory. In aversive memory formation, loss of Dop1R2 could lead to enhanced or impaired memory, depending on the activated signaling pathways and the internal

state of the animal. For reward memory formation, loss of Dop1R2 seems to impair memory. Moreover, Rac does not have an effect on reward memories (Yang et al., 2023 [DOI](#)). However, it still remains unclear, how all of these aspects are integrated and if there is a hierarchical order.

DANs can produce aversive and appetitive associations depending on the temporal presentation of odor cue and reinforcement stimulus (Handler et al., 2019 [DOI](#)). Thus, dopaminergic signaling is able to modify the KC-MBON synapses bi-directionally. The Dop1R1-G α_s -cAMP pathway seems to detect the temporal coincidence of the stimuli whereas the Dop1R2-G α_q -Ca²⁺ pathway detects the temporal ordering (Handler et al., 2019 [DOI](#)). Dop1R2 mutant flies seem to be able to form an odor association but are not able to update it. Further, Dop1R2 is required for the potentiation of the KC-MBON synapse following backward pairing.

Taken together, Dop1R2 has multiple roles during memory formation and integrates different signals, including the detection of the order of stimuli, the internal state, like energy levels and forgetting and maintenance signals.

The receptor does not seem to be required for STM but for later timepoints. Previous studies looking at the temporal requirement of the lobes (Güven-Ozkan & Davis, 2014 [DOI](#); Perisse et al., 2013 [DOI](#)) defined the γ -lobe to be responsible for memory acquisition (Blum et al., 2009 [DOI](#); Trannoy et al., 2011 [DOI](#); Zars et al., 2000 [DOI](#)). The α/β -lobe and its output is involved in LTM (Akmal et al., 2011 [DOI](#); Blum et al., 2009 [DOI](#); Cervantes-Sandoval et al., 2013 [DOI](#); Huang et al., 2012 [DOI](#); Ichinose et al., 2015 [DOI](#); Krashes & Waddell, 2008 [DOI](#); Trannoy et al., 2011 [DOI](#)). The function of the α'/β' -lobe seems to be LTM consolidation (Cervantes-Sandoval et al., 2013 [DOI](#); Krashes & Waddell, 2008 [DOI](#)). However, both lobes also seem to have a role in middle-term memory (Bouzaiane et al., 2015 [DOI](#); Scheunemann et al., 2012 [DOI](#); Shyu et al., 2019 [DOI](#); Turrel et al., 2022 [DOI](#)). As loss of Dop1R2 in the γ -lobe or the whole MB does not impair STM, we conclude that the receptor is not required for this memory type. However, Dop1R2 is expressed in the γ -lobe (Crocker et al., 2016 [DOI](#); Croset et al., 2018 [DOI](#)), so it might be required for other types of behaviors.

The impairment of reward LTM upon knock-out of Dop1R2 in the α/β -lobe and the α'/β' -lobe matches the described role of these neurons. So, both the α/β -lobe and the α'/β' -lobe require Dop1R2 for LTM. Interestingly, both lobes also seem to require Dop1R2 for two-hour memories. As the MB-MP1 DAN is active in this time window as well, this would suggest that Dop1R2 function at this time point is important for correct LTM formation.

This would indicate, that Dop1R1 is the main contributor to STM while Dop1R2 is responsible for later memory stages. It would be interesting to know how the switch from Dop1R1 dependency to Dop1R2 occurs.

As Dop1R2 is required in the MBON MVP2 for reward LTM, it would be exciting to see if it is also required in other MBONs or neurons outside the MB. Besides learning and memory, the MB also uses dopamine to regulate sleep. This tool offers the opportunity to study both of these aspects in neurons of interest.

The genetic tool we generated here to study the role of the Dop1R2 dopamine receptor in cells of interest, provides versatile possibilities as it can be used in combination with the powerful genetic tools of *Drosophila*. Using this line, we could show, that Dop1R2 is specifically required for later memory stages of both aversive and appetitive memory in the α/β -lobe and the α'/β' -lobe.

Acknowledgements

We would like to thank H. Tanimoto, the Kyoto and Bloomington stock centers for fly strains. We would like to thank colleagues of the Sprecher lab for their valuable input and discussions.

Author Contribution

Conceptualization: SGS, Data curation: JCK, Formal Analysis: JCK, Funding acquisition: SGS, Investigation: JCK, NK, CF, Methodology: JCK, CF, Project administration: SGS, Resources: SGS, Supervision: SGS, Validation: SGS, CF, Visualization: JCK, NK, Writing – original draft: JCK

Funding

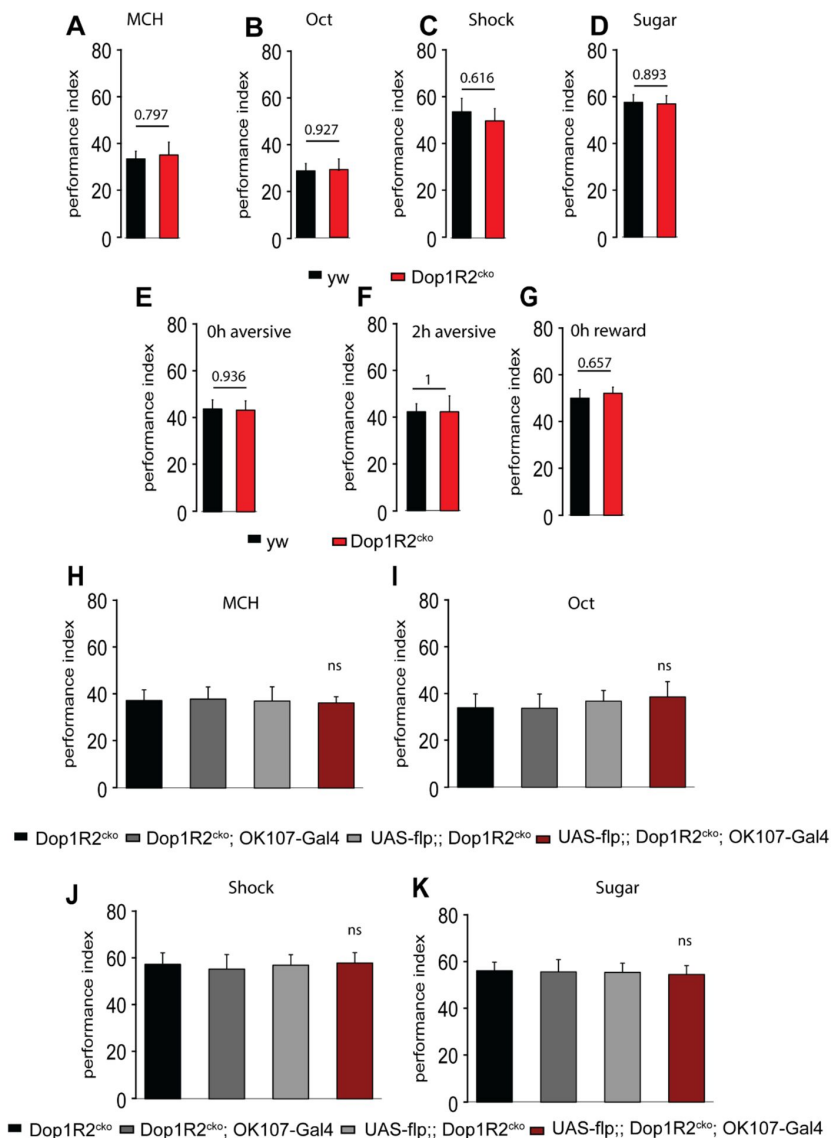
The current work is supported by the Swiss National Science Foundation and the Novartis foundation for Biomedical Research to SGS.

Declaration of Interests

The authors declare no competing interests.

Supplementary figure legends

S1. Sequences for cloning $Dop1R2^{cko}$. A) Sequence of *Dop1R2* with crispr sites. B) Sequence of the Donor plasmid with all three fragments of *Dop1R2*, the FRT sites and the HA-Tag



S2.

Sensory tests of the *dop1R2* conditional knock-out line.

A) Response of yw and *Dop1R2^{cko}* to MCH B) Response of yw and *Dop1R2^{cko}* to Oct C) Response of yw and *Dop1R2^{cko}* to Shock D) Response of yw and *Dop1R2^{cko}* to Sugar. E) 0h aversive memory of yw and *Dop1R2^{cko}* F) 0h aversive memory of yw and *Dop1R2^{cko}* G) 0h aversive memory of yw and *Dop1R2^{cko}*. In all tested conditions the *Dop1R2^{cko}* shows no significant difference to the control line. H-K) Sensory responses of flies with *Dop1R2* knock-out in the whole MB alongside the parental control lines. H) Response to MCH, I) Response to Oct, J) Response to Shock, K) Response to Sugar. Loss of *Dop1R2* in the whole MB does not affect sensory responses. See S1 Table for the data. Bar graphs represent the mean, and error bars represent the standard error of the mean. For each shown graph the N = 12. Asterisks denote significant differences between groups (*p < 0.05, **p < 0.005, ***p < 0.001, ns: p > 0.05).

Tab 1. Performance score of all the learning assays.

Methods

Lead Contact and Data Availability

Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Simon Sprecher (simon.sprecher@unifr.ch). All used data is in this manuscript.

Fly Husbandry

Drosophila melanogaster flies were reared in plastic vials on standard cornmeal food (12g Agar, 40g Sugar, 40g Yeast, 80g Cornmeal per Liter) and transferred to fresh food vials every 2-3 days. Flies were generally kept at 25°C, 60-65% humidity, and exposed to 12 h light and 12 h darkness with light onset at 8 am. The following fly lines were used: y[1] w[*] P{y[+t7.7]=nos-phiC31\int.NLS}X; P{y[+t7.7]=CaryIP}su(Hw)attP6 (abbreviated *nos>Cas9* in this paper; BL 32232) for microinjection and as PCR template, w;; Dr e/TM3 (BL 36305) was used as 3rd chromosomal balancer line. w[1118], 20XUAS-FLPG5.PEST (BL 55807) was used as UAS-flp. OK107-Gal4 (106098) was obtained from the Kyoto stock center. The 5HTR1B-Gal4 line (BL 27636) and the c305a-Gal4 (BL 30829) line are from Bloomington stock center. The c739-Gal4 line was gifted to us by H. Tanimoto (Tohoku University Japan). The yw (BL 1495) line was used as the control line.

Generation of dop1R2^{cko}

For generating the conditional knock-out line we needed three regions: 1) The 5' flanking sequence upstream of the first FRT site as homologous region, 2) The 3' flanking sequence downstream of the second FRT site as homologous region, 3) the sequence in between the two FRT sites hereafter named Dop1R2 coding fragment. To obtain the dopamine receptor fragments, genomic DNA from *nos>Cas9* flies was used as the template for the PCRs. The primers used for the different PCR fragments are shown in **Table 2**. The 3' and 5' fragments of Dop1R2 were sub-cloned into pBluescript II SK(+) vector (pBS) with adequate restriction enzymes – SpeI and SmaI for the 5' fragment and XhoI and Acc65I for the 3' fragment. The Dop1R2 coding fragment was cloned with the Invitrogen TOPO kit. After sequence confirmation the fragments were cloned subsequently into pBS-FRT-3xHA-FRT. The vector is modified from the vector used by Widmer et al. (Widmer et al., 2018). The GFP was replaced by 3xHA using 3xHA Nco fw
cacatgggtTACCCATACGATGTTCTGACTATGCGGGCTATCCCTATGACGTCCCC
GACTATGCAGGATCCTATCCATATGACGTTCCAGATTACGCTgca and 3xHA H3 re
agcttgacAGCGTAATCTGGAACGTCATATGGATAGGATCCTGCATAGTCCGGGACG
TCATAGGGATAGCCCGCATAGTCAGGAACATCGTATGGGTac as oligos whereas the FRT site and the pBS backbone were kept. The GPBS fragment was cloned in-frame in front of the 3xHA tag using AgeI and BstEII-HF as restriction enzymes.

The two guide RNAs were chosen using crispr optimal target finder (<http://targetfinder.flycrispr.neuro.brown.edu/>). The used sequences (**Table 1**) were integrated into PCR primers to amplify a fragment of the pCFD4-U6_1_6_3tandemgRNAs plasmid (a gift from Simon Bullock, addgene plasmid # 49411 (Port et al., 2014)) that was cloned into BbsI digested pCFD4-U6_1_6_3tandemgRNAs vector using Gibson assembly (NEB). Insertion of the gRNAs was confirmed by sequencing. A mix containing 0.4 mg/ml template and 0.2 mg/ml of the guide RNAs plasmid was injected into *nos>Cas9* embryos. The injected flies were crossed with a 3rd chromosomal balancer line. The F1 generation was crossed again with the 3rd chromosomal balancer flies. As soon as eggs or larvae were visible the adults were sacrificed to check for the *Dop1R2* construct via PCR. Positive hits were then sequenced, and a stock was established.

Key Resources Table				
Reagent type (species) or resource	Designation	Source or reference	Identifiers	Additional information
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	y[1] P{y[+t7.7]=nos-phiC31\int.NLS}X; P{y[+t7.7]=CaryIP}su(Hw)attP6 w[*]	Bloomington Stock Center	RRID:BDSC_32232	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	w;; Dr e/TM3	Bloomington Stock Center	RRID:BDSC_36305	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	w[1118], 20XUAS-FLPG5.PEST	Bloomington Stock Center	RRID:BDSC_55807	
Genetic Reagent <i>Drosophila</i>	OK107-Gal4	Kyoto Stock Center	DGGR_106098	

Table 1.
 Key Resource Table

Table 1. (continued)

<i>melanogaster</i> both male and female				
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	5HTR1B-Gal4	Bloomington Stock Center	RRID:BDSC_27636	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	c739-Gal4	Hiromu Tanimoto (Tohoku University Japan)	RRID:BDSC_7362	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	c305a-Gal4	Bloomington Stock Center	RRID:BDSC_30829	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	yw	Bloomington Stock Center	RRID:BDSC_1495	
Genetic Reagent <i>Drosophila melanogaster</i> both male and female	FRT-Dop1R2-HA-FRT Dop1R2 ^{cko}	This paper		See Materials and Methods section: Creation of Dop1R2 ^{cko}
antibody	Mouse monoclonal α -HA clone 12CA5	Roche	RRID:AB_514505 11583816001	1:200
antibody	Rat monoclonal α -Dros-N-cadherin (Ncad)	Developmental Studies Hybridoma Bank	RRID:AB_528121 DN-Ex #8	1:30
Plasmid	pCFD4-U6_1_6_3tandemgRNAs plasmid	Simon Bullock	RRID:Addgene_49411	

Table 2

Primers and gRNAs for generating Dop1R2 conditional knock-out lines.

	Primer Name	Sequence
Dop1R2		
Cloning Primers	Dop1R2_5' fw_SpeI	GCTGCAG ACTAGT CAGCCACCACA
	Dop1R2_5' re_SmaI	CCTGA ACCCGGG GATAAACTTTACCATAATGC
	Dop1R2_fragment_fw_AgeI	GTTCG ACCGGT GATTGCATTGTGTTCAACAG
	Dop1R2_fragment re_BstEII	GGGCTT GGTAACC ACGACGAATCTTGCGTGGACAG
	Dop1R2_3' fw_XhoI	ATT CTCGAGA AGTATCAACCCACGATGCGTTC
	Dop1R2_3' re_Acc65I	GGCAAC GGTACC AGATGCAGATACCG
gRNAs	Dop1R2_gRNA_5'	ACGAACTTAAGATAAAGTGTCGG
	Dop1R2_gRNA_3'	GCATCGTGGGCTGGTACTTCCGG
Screening & Sequencing	Dop1R2_Screening_for	TATCCCTATGACGTCCCGGAC
	Dop1R2_Screening_re	GACAGGTTGAGTGATGCGCC
	Dop1R2_5'FRT_for	GGCTACACATCATTTTATGCCAG
	Dop1R2_5'FRT_re	GTTCTGTGCCTGATTCTGTTC
	Dop1R2_3'FRT_for	TCCTAACTGGCTTCTCTTCC
	Dop1R2_3'FRT_re	AGCGCTTAATTACGAAAGC

Immunostainings

Brains from 5–8-day old flies were dissected in PBS (Bio-Froxx 1346LT050) and fixed in 3.7% formaldehyde for 25 min at RT. The brains were washed with 1X PBS containing 0.5% Triton X-100 (Carl Roth 3051.3) (0.5% PBST) before incubating the primary antibodies o/n at 4°C. The primary antibodies were mouse α -HA clone 12CA5 (Roche) 1:200, α -Drosophila-cadherin (Ncad) rat (Iowa H.B: DN-EX8) 1:30. After the brains were washed in 0.5% PBST again, they were incubated o/n at 4°C with the secondary antibodies. The following secondary antibodies were used: Goat α -rat Alexa 647 (Molecular probes A-21247) and Goat α -Mouse Alexa 488 (Molecular probes A11029) 1:200. The brains were washed again and mounted in self-made mounting media (90% Glycerol (Fischer Scientific Catalog No. BP229-1), 0.5% N-propyl gallate (Sigma P3130), 20 mM Tris (Fischer Scientific, Catalog No. BP152-5), pH 8.0) (Adapted from NIC Harvard Medical School). The brains were imaged using a confocal microscope (Leica STELLARIS 8 FALCON) at 40X magnification with the Plan APO 40x/1.10 water immersion objective at 1,024 × 1,024 pixels resolution and 600Hz scan rate. Images were processed with Fiji ImageJ and Adobe Illustrator.

Learning Apparatus

For behavior experiments, we used a memory apparatus that is based on Tully and Quinn's design and modified to allow conducting four memory experiments in parallel (CON-Elektronik, Greussenheim, Germany). Experiments were performed at 23–25°C and 65–75% relative humidity. The training was performed in dim red light and memory tests were done in complete darkness. The two odors used were 3-octanol (3-Oct) (Sigma-Aldrich 218405) and 4-methyl-cyclohexanol (MCH) (Sigma-Aldrich 66360) diluted in paraffin oil (Sigma-Aldrich 18512) 1:100 respectively. 260 μ l of the diluted odors were presented in a plastic cup of 14 mm in diameter. A vacuum membrane pump ensured odor delivery at a flow rate of 7 l/min.

Aversive olfactory conditioning

For aversive conditioning, groups of 50–100 flies of mixed sex were loaded in tubes lined with an electrifiable copper grid. The position in the machine and the sequence in which the genotypes were tested were randomized. Experiments in which more than half of the flies died, the flies did not move or there were technical problems with the machine, as well as human errors were excluded. The training was conducted in the morning. After an accommodation period of 90 s, the first odor was presented for 60 s. In parallel, 12 pulses of 100 V for 1.5 s were delivered with an interval of 3.5 s. After 30 s of flushing with fresh air, the second odor was presented for 60 s. For the subsequent group of flies, the order of the two odors was reversed. For measuring 0 h performance flies were tested about 3 minutes after the end of the conditioning. To determine 2h memory performance, flies were transferred to food vials after conditioning and kept at 25°C until the test. For each genotype and condition the biological replicate is N=12.

Appetitive olfactory conditioning

Before appetitive conditioning, groups of 50 to 100 flies with mixed sex were starved for 19 to 21 h in plastic vials containing damp cotton at the bottom. Experiments in which more than half of the flies died, the flies did not move or there were technical problems with the machine, as well as human errors were excluded. The position in the machine and the sequence in which the genotypes were tested were randomized. The training was conducted in the morning. The conditioning protocol consists of a 90-s accommodation period, 120 s of the first odor, 60 s of fresh air followed by 120 s of the second odor. During the first odor, flies are in a conditioning tube lined with filter paper that was soaked in water the day before the experiment and left to dry overnight. For the second odor, flies are transferred to a conditioning tube lined with a filter paper that was soaked with a 1.5 M sucrose (Sigma-Aldrich, Cat# 84100-1KG; CAS Number 57-50-1) solution on the day before and left to dry at RT. After conditioning, flies were either directly tested for STM or put

back in starvation vials until the memory test 2 h later. For 24-h memory, flies were fed for 3 h after training before starving them again. One experiment consisted of 2 reciprocal conditionings, in which the odor paired with sucrose was reversed. For each genotype and condition the biological replicate is N=12.

Memory tests

Flies were loaded into a sliding compartment and transferred to a two-arm choice point. Animals were allowed to choose between 3-octanol and 4-methyl-cyclohexanol. After 60 s, flies trapped in both arms were collected separately and counted. Based on these numbers, a preference index was calculated as follows:

$PREF = ((N_{arm1} - N_{arm2}) / N_{total}) \cdot 100$ the two preference indices were calculated from the two reciprocal experiments. The average of these two PREFs gives a memory performance index (PI).
 $PI = (PREF_1 + PREF_2) / 2$

Sensory Accuracy tests

Flies were tested for their ability to sense the two used odors 3-octanol and 4-methyl-cyclohexanol as well as electric shock. Therefore, the flies were loaded into a sliding compartment and brought to a two-arm choice point. The flies were allowed to freely choose between an arm containing the stimulus and a neutral arm. All experiments were carried out in the dark. Afterward, the flies in each arm were counted and a preference index was calculated.

For testing the odor response, the flies could choose between one of the odors in the same concentration as used for the behavior experiment and the same amount of paraffin oil for 120s.

Preference index $PI = ((N_{air} - N_{odor}) / N_{total}) \cdot 100$.

For shock response, the flies could freely choose for 60s between a cooper-grid lined tube getting pulses of 100V or a cooper-grid lined tube getting no electric shock. Preference index $PI = ((N_{No\ shock} - N_{shock}) / N_{total}) \cdot 100$.

For testing sugar sensitivity, a group of flies was starved for 1 to 21 h in a tube with damp cotton on the bottom. They could choose for 120 s between a tube lined with filter paper that was soaked in 1.5 M sucrose solution the day before or a tube lined with filter paper that was soaked in distilled water the day before. Preference index $PI = ((N_{sucrose} - N_{water}) / N_{total}) \cdot 100$.

Statistical analysis

To compare performance indices between different groups we used One-way ANOVA (ANalysis Of VAriance) with posthoc Tukey HSD (Honestly Significant Difference) Test Calculator for comparing multiple treatments in R with the package multcomp. In the case of two groups, we performed a *t*-test for comparison.

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Reviewer #1 (Public Review):

Summary:

In this manuscript, the authors present a novel CRISPR/Cas9-based genetic tool for the dopamine receptor dop1R2. Based on the known function of the receptor in learning and memory, they tested the efficacy of the genetic tool by knocking out the receptor specifically in mushroom body neurons. The data suggest that dop1R2 is necessary for longer-lasting memories through its action on α/β and α'/β' neurons but is dispensable for short-term memory and thus in γ neurons. The experiments impressively demonstrate the value of such a genetic tool and illustrate the specific function of the receptor in subpopulations of KCs for longer-term memories. The data presented in this manuscript are significant.

<https://doi.org/10.7554/eLife.99368.1.sa2>

Reviewer #2 (Public Review):

Summary:

This manuscript examines the role of the dopamine receptor, Dop1R2, in memory formation. This receptor has complex roles in supporting different stages of memory, and the neural mechanisms for these functions are poorly understood. The authors are able to localize

Dop1R2 function to the vertical lobes of the mushroom body, revealing a role in later (presumably middle-term) aversive and appetitive memory. In general, the experimental design is rigorous, and statistics are appropriately applied. While the manuscript provides a useful tool, it would be strengthened further by additional mechanistic studies that build on the rich literature examining the roles of dopamine signaling in memory formation. The claim that Dop1R2 is involved in memory formation is strongly supported by the data presented, and this manuscript adds to a growing literature revealing that dopamine is a critical regulator of olfactory memory. However, the manuscript does not necessarily extend much beyond our understanding of Dop1R2 in memory formation, and future work will be needed to fully characterize this reagent and define the role of Dop1R2 in memory.

Strengths:

- (1) The FRT lines generated provide a novel tool for temporal and spatially precise manipulation of Dop1R2 function. This tool will be valuable to study the role of Dop1R2 in memory and other behaviors potentially regulated by this gene.
- (2) Given the highly conserved role of Dop1R2 in memory and other processes, these findings have a high potential to translate to vertebrate species.

Weaknesses:

- (1) The authors state Dop1R2 associates with two different G-proteins. It would be useful to know which one is mediating the loss of aversive and appetitive memory in Dop1R2 knockout flies.
- (2) It would be interesting to examine 24hr aversive memory, in addition to 24hr appetitive memory.
- (3) The manuscript would be strengthened by added functional analysis. What are the DANs that signal through Dop1R. How do these knockouts impact MBONs?
- (4) Also in Figure 2, the lobe-specific knockouts might be moved to supplemental since there is no effect. Instead, consider moving the control sensory tests into the main figure.
- (5) Can the single-cell atlas data be used to narrow down the cell types in the vertical lobes that express Dop1R2? Is it all or just a subset?

<https://doi.org/10.7554/eLife.99368.1.sa1>

Reviewer #3 (Public Review):

Summary:

Kaldun et al. investigated the role of Dopamine Receptor Dop1R2 in different types and stages of olfactory associative memory in *Drosophila melanogaster*. Dop1R2 is a type 1 Dopamine receptor that can act both through Gs-cAMP and Gq-ERCa²⁺ pathways. The authors first developed a very useful tool, where tissue-specific knock-out mutants can be generated, using Crispr/Cas9 technology in combination with the powerful Gal4/UAS gene-expression toolkit, very common in fruit flies.

They direct the K.O. mutation to intrinsic neurons of the main associative memory centre fly brain-the mushroom body (MB). There are three main types of MB-neurons, or Kenyon cells, according to their axonal projections: a/b; a'/b', and g neurons.

Kaldun et al. found that flies lacking dop1R2 all over the MB displayed impaired appetitive middle-term (2h) and long-term (24h) memory, whereas appetitive short-term memory

remained intact. Knocking-out dop1R2 in the three MB neuron subtypes also impaired middle-term, but not short-term, aversive memory.

These memory defects were recapitulated when the loss of the dop1R2 gene was restricted to either a/b or a'/b', but not when the loss of the gene was restricted to g neurons, showcasing a compartmentalized role of Dop1R2 in specific neuronal subtypes of the main memory centre of the fly brain for the expression of middle and long-term memories.

Strengths:

(1) The conclusions of this paper are very well supported by the data, and the authors systematically addressed the requirement of a very interesting type of dopamine receptor in both appetitive and aversive memories. These findings are important for the fields of learning and memory and dopaminergic neuromodulation among others. The evidence in the literature so far was generated in different labs, each using different tools (mutants, RNAi knockdowns driven in different developmental stages...), different time points (short, middle, and long-term memory), different types of memories (Anesthesia resistant, which is a type of protein synthesis independent consolidated memory; anesthesia sensitive, which is a type of protein synthesis-dependent consolidated memory; aversive memory; appetitive memory...) and different behavioral paradigms. A study like this one allows for direct comparison of the results, and generalized observations.

(2) Additionally, Kaldun and collaborators addressed the requirement of different types of Kenyon cells, that have been classically involved in different memory stages: g KCs for memory acquisition and a/b or a'/b' for later memory phases. This systematical approach has not been performed before.

(3) Importantly, the authors of this paper produced a tool to generate tissue-specific knock-out mutants of dop1R2. Although this is not the first time that the requirement of this gene in different memory phases has been studied, the tools used here represent the most sophisticated genetic approach to induce a loss of function phenotypes exclusively in MB neurons.

Weaknesses:

(1) Although the paper does have important strengths, the main weakness of this work is that the advancement in the field could be considered incremental: the main findings of the manuscript had been reported before by several groups, using tissue-specific conditional knockdowns through interference RNAi. The requirement of Dop1R2 in MB for middle-term and long-term memories has been shown both for appetitive (Musso et al 2015, Sun et al 2020) and aversive associations (Plačais et al 2017).

(2) The approach used here to genetically modify memory neurons is not temporally restricted. Considering the role of dopamine in the correct development of the nervous system, one must consider the possible effects that this manipulation can have in the establishment of memory circuits. However, previous studies addressing this question restricted the manipulation of Dop1R2 expression to adulthood, leading to the same findings than the ones reported in this paper for both aversive and appetitive memories, which solidifies the findings of this paper.

(3) The authors state that they aim to resolve disparities of findings in the field regarding the specific role of Dop1R2 in memory, offering a potent tool to generate mutants and addressing systematically their effects on different types of memory. Their results support the role of this receptor in the expression of long-term memories, however in the experiments performed here do not address temporal resolution of the genetic manipulations that could bring light into the mechanisms of action of Dop1R2 in memory. Several hypotheses have been proposed,

from stabilization of memory, effects on forgetting, or integration of sequences of events (sensory experiences and dopamine release).

Overall, the authors generated a very useful tool to study dopamine neuromodulation in any given circuit when used in combination with the powerful genetic toolkit available in *Drosophila*. The reports in this paper confirmed a previously described role of Dop1R2 in the expression of aversive and appetitive LTM and mapped these effects to two specific types of memory neurons in the fly brain, previously implicated in the expression and consolidation of long-term associative memories.

<https://doi.org/10.7554/eLife.99368.1.sa0>